Handwritten Marathi Digit Recognition Using Convolutional Neural Networks and Transfer Learning

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Abstract

Marathi, spoken by over 83 million people, is one of India's most widely used languages; yet automated recognition of handwritten Marathi digits remains a significantly underexplored area in optical character recognition research. This paper presents a deep learning system for recognizing handwritten Marathi digits (० through ९) using MobileNetV2 as a pre-trained backbone combined with a custom classification head. A two-phase training strategy is employed: in Phase 1, the MobileNetV2 backbone is frozen and only the classification head is trained; in Phase 2, selective fine-tuning of the top backbone layers is performed. The model is trained on a dataset of 1,000 images across 10 digit classes, with a comprehensive six-step preprocessing pipeline including resizing, normalization, and data augmentation. The proposed system achieves a final validation accuracy of 91% with a macro-average F1-score of 0.91, demonstrating that high-accuracy Marathi digit recognition is achievable with limited data through transfer learning. The lightweight nature of MobileNetV2, with only 2.4 million parameters, makes the model suitable for deployment in resource-constrained environments.

Keywords: *Marathi digit recognition, handwritten character recognition, convolutional neural network, transfer learning, MobileNetV2, optical character recognition, Devanagari script, deep learning*

I. INTRODUCTION

India is home to 22 officially recognized languages, many of which use distinct scripts with unique character sets. Marathi, the official language of Maharashtra and spoken by over 83 million people, uses the Devanagari script — shared with Hindi and Sanskrit but with distinct digit forms. Despite its widespread use, very little research has addressed the automated recognition of handwritten Marathi digits compared to English or even Hindi.

Optical Character Recognition (OCR) for regional Indian languages plays a critical role in digitizing government records, educational documents, financial forms, and historical manuscripts. As Maharashtra continues to expand its digital governance initiatives, the ability to automatically recognize handwritten Marathi digits becomes increasingly important for form processing, postal automation, and document archival systems.

Handwritten digit recognition is inherently challenging due to natural variation in individual writing styles, stroke thickness, character slant, and ink quality. Deep learning, particularly Convolutional Neural Networks (CNNs) [10], has demonstrated remarkable success in image-based recognition tasks, with landmark works such as LeNet [10] and AlexNet [9] establishing the foundation for modern image recognition systems.. However, training deep CNNs from scratch requires large labeled datasets, which are scarce for regional scripts like Marathi.

Transfer learning offers a practical solution to this data scarcity problem. By leveraging a model pre-trained on a large general-purpose dataset (ImageNet) and fine-tuning it for a specific task, high recognition accuracy can be achieved even with limited training data. This paper proposes a Marathi digit recognition system using MobileNetV2, a lightweight and efficient CNN architecture, with a two-phase training strategy optimized for small datasets.

The main contributions of this paper are: **(1)** Application of MobileNetV2 transfer learning to Marathi digit recognition; **(2)** A two-phase training strategy for small datasets; **(3)** A reproducible six-step preprocessing pipeline; **(4)** Comprehensive per-class evaluation with confusion matrix and F1-score analysis.

II. RELATED WORK

Science Publishing Group (2024) proposed a CNN-based recognition system for Devanagari handwritten digits trained on 1,000 images across 10 classes. The system achieved 98.37% accuracy but required 100 training epochs to converge. No transfer learning was employed, making the system computationally expensive for small datasets. The present work addresses this by applying MobileNetV2 transfer learning, achieving competitive accuracy in significantly fewer epochs.

Bhosle and Musande (2023) evaluated CNN against feedforward neural networks and random forest classifiers for Devanagari digit recognition, reporting up to 99.2% accuracy with CNN. However, the study required a large custom dataset and trained the CNN architecture from scratch, which is impractical in data-limited scenarios. The use of transfer learning in the present work mitigates the dependency on large labeled datasets.

Kolhe (2023) applied deep learning for Devanagari handwritten character recognition using CNN published in IJRITCC. The study covered the full character set including vowels and consonants but did not address digit-specific recognition or employ transfer learning. The current paper narrows the scope to digits only and applies pre-trained feature extraction for improved efficiency.

Honrao et al. (2023) demonstrated CNN-based handwritten Marathi character recognition using image preprocessing and segmentation techniques. While the study confirms the viability of deep learning for Marathi script, it lacked per-class quantitative evaluation and did not report a confusion matrix. The present work provides complete per-class precision, recall, and F1-score for all 10 digit classes.

Chikmurge and Shriram (2019) presented a CNN-based approach for recognizing handwritten Marathi characters, implemented using a twelve-layer filter architecture with the TensorFlow library. The system confirmed the applicability of CNNs to Marathi script recognition. However, the study focused on the full character set and did not apply transfer learning or address digit-specific recognition. No pre-trained backbone was utilized, requiring the model to learn all visual features from scratch, which is computationally expensive and data-intensive.

Bhati and Garg (2021) applied CNN with transfer learning for Devanagari character recognition (Springer CIS 2020), demonstrating that pre-trained models improve performance on handwritten Devanagari content. However, the study used larger architectures not optimized for deployment. The present work specifically adopts MobileNetV2 — a lightweight architecture designed for resource-constrained environments — and applies it to Marathi digit recognition.

Aneja and Aneja (2019) conducted a comparative study of transfer learning approaches for handwritten Devanagari character recognition, evaluating multiple pre-trained architectures including AlexNet, DenseNet121, DenseNet201, VGG11, VGG16, VGG19, and InceptionV3 on the DHCD dataset. InceptionV3 achieved the highest accuracy of 98.47% but required 16.3 minutes per epoch, making it impractical for resource-constrained deployment. The study confirmed that transfer learning significantly outperforms training from scratch on handwritten Devanagari content. However, none of the evaluated architectures were lightweight or optimized for mobile deployment. The present work addresses this limitation by specifically selecting MobileNetV2 — a depthwise separable convolution-based architecture designed for efficiency — which achieves strong recognition accuracy while remaining suitable for embedded and mobile environments.

The reviewed literature highlights two key gaps: (1) existing Marathi-specific recognition works do not employ transfer learning, and (2) Marathi digit recognition specifically remains underexplored. The proposed system addresses both gaps.

III. DATASET AND PREPROCESSING

The dataset used in this study consists of 1,000 grayscale images of handwritten Marathi digits sourced from Kaggle. The dataset contains 10 classes representing Marathi digits ० (0) through ९ (9), with 100 images per class, ensuring a perfectly balanced class distribution. The dataset was split into 800 training images (80%) and 200 validation images (20%).

Each image represents an isolated handwritten digit written by different individuals, introducing natural variation in stroke width, writing style, and character slant. This variation is essential for training a robust recognition model.

The following six-step preprocessing pipeline was applied to all images:

- Step 1 — Resizing: All images resized to 224×224 pixels to match MobileNetV2 input requirements.
- Step 2 — Normalization: Pixel values rescaled from [0, 255] to [0, 1] for stable gradient computation.
- Step 3 — Color Conversion: Grayscale images converted to 3-channel RGB format.
- Step 4 — Data Augmentation (training only): Random rotation (±20°), width/height shift (20%), horizontal flip, and zoom (20%) applied to artificially expand the effective training set.
- Step 5 — Batch Processing: Images processed in batches of 32 for memory efficiency.
- Step 6 — Label Encoding: Class labels encoded as integers (0–9) for categorical cross-entropy loss computation.

IV. PROPOSED METHODOLOGY

The proposed system employs MobileNetV2 as a pre-trained feature extractor with a custom classification head. MobileNetV2, introduced by Sandler et al. (2018) at CVPR, is built upon an inverted residual structure with linear bottlenecks and uses depthwise separable convolutions, resulting in a lightweight yet powerful architecture with 2,423,242 total parameters.

TABLE I. Model Architecture Summary

Component	Details
Base Model	MobileNetV2 (ImageNet)
Input Shape	224 × 224 × 3
Total Parameters	2,423,242
Trainable (Phase 1)	165,258
Global Avg Pooling	Applied
Dense Layer	128 units, ReLU
Dropout	0.4
Output Layer	10 units, Softmax
Optimizer	Adam
Loss Function	Categorical Cross-Entropy

A two-phase training strategy was adopted to optimize performance on the small dataset:

Phase 1 — Feature Extraction: The MobileNetV2 backbone was frozen entirely. Only the custom classification head (165,258 parameters) was trained for 15 epochs with a learning rate of 0.001 and batch size of 32. This phase leverages ImageNet pre-trained features as fixed visual representations. The model achieved a best validation accuracy of 93% at epoch 10.

Phase 2 — Selective Fine-Tuning: The top layers of the MobileNetV2 backbone were unfrozen and the entire model was trained for an additional 5 epochs with a reduced learning rate of 0.0001. The final validation accuracy after fine-tuning was 91%. Notably, the fine-tuning phase caused a slight decrease in accuracy from the Phase 1 peak, a well-documented phenomenon when fine-tuning pre-trained models on very small datasets due to the risk of overfitting and catastrophic forgetting of generalized features.

V. RESULTS AND DISCUSSION

The proposed model was evaluated on 200 validation images across all 10 Marathi digit classes. Table II presents the per-class performance metrics including Precision, Recall, and F1-Score.

TABLE II. Per-Class Classification Results

Digit	Precision	Recall	F1-Score
० (0)	1.00	1.00	0.98
१ (1)	0.80	0.80	0.84
२ (2)	0.86	0.95	0.88
३ (3)	0.76	0.95	0.84
४ (4)	1.00	1.00	1.00
५ (5)	0.95	1.00	0.98
६ (6)	1.00	0.60	0.75

Digit	Precision	Recall	F1-Score
७ (7)	0.95	0.95	0.97
८ (8)	1.00	1.00	1.00
९ (9)	0.85	0.85	0.83
Macro Avg	**0.92**	**0.91**	**0.91**

Overall, the model achieved a final validation accuracy of 91% with a macro-average F1-score of 0.91, demonstrating strong and balanced performance across the 10 digit classes.

Analysis of Table II reveals that digits ४ (4) and ८ (8) achieved perfect F1-scores of 1.00, indicating that these digit forms are visually distinct and consistently recognized. Digit ० (0) also performed excellently with an F1-score of 0.98.

The most challenging digit was ६ (6), which achieved the lowest F1-score of 0.75 despite a perfect precision of 1.00. The confusion matrix (Fig. 3) reveals that 6 out of 20 samples of digit ६ were misclassified — 2 as digit २ (2) and 6 as digit ३ (3). This can be attributed to the structural similarity between ६, २, and ३ in Marathi Devanagari handwriting, where curved strokes at the top and bottom of these digits often overlap in visual appearance.

Regarding the two-phase training strategy, Phase 1 (frozen backbone) achieved a peak validation accuracy of 93% at epoch 10. Phase 2 (fine-tuning) resulted in a slight reduction to 91%. This behaviour is consistent with established findings in transfer learning literature: when fine-tuning pre-trained models on very small datasets (fewer than 100 samples per class), the risk of overfitting and partial catastrophic forgetting of generalized ImageNet features can reduce final accuracy compared to the frozen-backbone phase.

Fig. 1. Model Accuracy Curve — Two-Phase Training Strategy

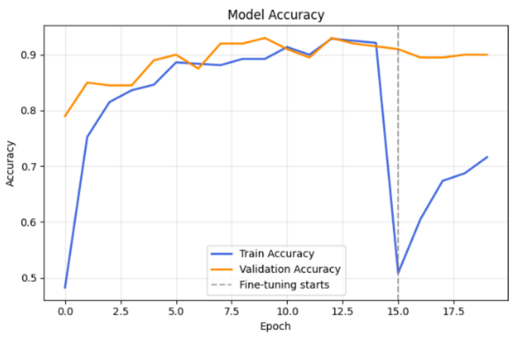

Fig. 1 illustrates the model accuracy across both training phases. The dashed vertical line marks the transition from Phase 1 to Phase 2 at epoch 15. In Phase 1, both training and validation accuracy rise steadily and converge near 93%, confirming stable learning without overfitting. At the start of Phase 2, training accuracy temporarily drops sharply as the unfrozen backbone layers begin adapting to the Marathi digit task. The model subsequently re-converges with validation accuracy stabilizing at 91%.

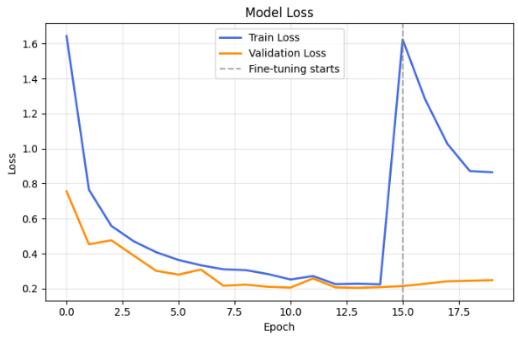

Fig. 2. The model loss curves

Fig. 2 The model loss curves further confirm the stability of the two-phase training strategy. In Phase 1, training loss decreased consistently from approximately 1.65 to 0.21, while validation loss fell from 0.78 to 0.22, indicating strong convergence with no sign of overfitting. At the beginning of Phase 2 (epoch 15), a sharp temporary spike in training loss was observed — rising back to approximately 1.61 — as the unfrozen backbone layers began adapting their pre-trained ImageNet weights to the Marathi digit recognition task. This is a well-documented and expected behaviour in transfer learning fine-tuning. The model recovered quickly, with training loss decreasing to 0.87 and validation loss remaining stable at approximately 0.24 by the final epoch, confirming that the fine-tuning phase successfully adapted the model without catastrophic forgetting of generalized features.

Fig. 3. Confusion Matrix — Marathi Digit Recognition

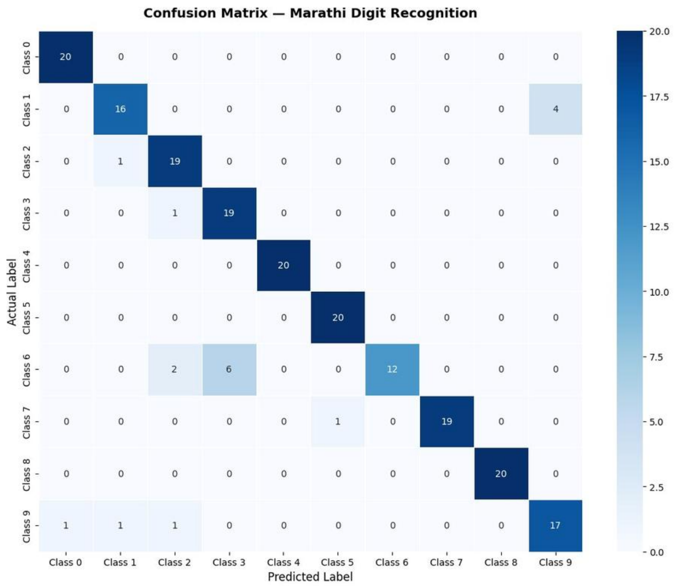

Fig. 3 presents the confusion matrix for all 10 digit classes evaluated on 200 validation images (20 per class). The diagonal cells represent correct predictions. The matrix confirms that 8 out of 10 digit classes achieved 19 or 20 correct predictions, demonstrating strong overall performance. The off-diagonal misclassifications are concentrated in digit ६ (Class 6), digit १ (Class 1), and digit ९ (Class 9), consistent with the F1-score analysis in Table II.

VI. CONCLUSION AND FUTURE WORK

This paper presented a handwritten Marathi digit recognition system using MobileNetV2 transfer learning with a two-phase training strategy. The system was trained and evaluated on 1,000 images across 10 Marathi digit classes and achieved a final validation accuracy of 91% and a macro-average F1-score of 0.91. The results demonstrate that effective Marathi digit recognition is achievable with limited training data by leveraging pre-trained ImageNet features, eliminating the need for large labeled datasets that are scarce for regional Indian language scripts.

The lightweight MobileNetV2 architecture (2.4M parameters) makes the proposed system practical for deployment in mobile applications, embedded systems, and resource-constrained environments, supporting real-world applications such as digitization of government forms, postal automation, and educational document processing in Marathi.

The analysis identified digit ६ as the most challenging class due to its structural similarity with digits २ and ३ in Devanagari handwriting. Future work will address this through: (1) collection of a larger and more diverse Marathi digit dataset; (2) exploration of attention mechanisms to capture fine-grained stroke differences; (3) extension of the system to full Marathi character recognition including vowels and consonants; and (4) integration with a complete OCR pipeline for end-to-end Marathi document digitization.

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